Light

GCSE Computer Science – Network Protocols (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a network protocol?

- ○ **A.** A type of network cable
- ○ **B.** A piece of hardware that connects networks
- ○ **C.** A type of malware that attacks networks
- ○ **D.** A set of rules that govern how data is transmitted across a network

Q2 (1 mark)

What does TCP/IP stand for?

- ○ **A.** Transmission Control Protocol/Internet Protocol
- ○ **B.** Technical Computer Program/Internet Process
- ○ **C.** Transfer Connection Protocol/Internal Program
- ○ **D.** Total Communication Package/Internet Provider

Q3 (1 mark)

What is HTTP used for?

- ○ **A.** Sending and receiving emails
- ○ **B.** Transferring web pages from web servers to browsers
- ○ **C.** Transferring files between computers
- ○ **D.** Encrypting data on a network

Q4 (1 mark)

What does the 'S' in HTTPS stand for?

- ○ **A.** Speed
- ○ **B.** Secure
- ○ **C.** Server
- ○ **D.** Standard

Q5 (1 mark)

What is FTP primarily used for?

- ○ **A.** Browsing websites
- ○ **B.** Transferring files between computers over a network
- ○ **C.** Sending emails
- ○ **D.** Making video calls

Q6 (1 mark)

Which protocol is used for sending emails?

- ○ **A.** HTTP
- ○ **B.** FTP
- ○ **C.** SMTP
- ○ **D.** POP

Q7 (1 mark)

What does POP stand for in email protocols?

- ○ **A.** Post Office Protocol
- ○ **B.** Personal Online Provider
- ○ **C.** Private Outgoing Post

- ○ **D.** Public Open Protocol

Q8 (1 mark)

What is a key feature of IMAP compared to POP?

- ○ **A.** IMAP is faster than POP
- ○ **B.** IMAP can only be used for sending emails
- ○ **C.** IMAP does not require an internet connection
- ○ **D.** IMAP keeps emails on the server, allowing access from multiple devices

Q9 (1 mark)

Why are network protocols organised into layers?

- ○ **A.** To separate different functions, making protocols easier to develop and update
- ○ **B.** To make networks physically taller
- ○ **C.** To make networks more expensive
- ○ **D.** To slow down data transmission

Q10 (1 mark)

Why are network standards important?

- ○ **A.** They allow devices from different manufacturers to communicate with each other
- ○ **B.** They make networks slower
- ○ **C.** They increase the cost of networking equipment
- ○ **D.** They prevent any communication between devices

Q11 (1 mark)

What is the purpose of encryption in network communications?

- ○ **A.** To make data travel faster
- ○ **B.** To compress data to save storage space
- ○ **C.** To protect data by converting it into a format that cannot be read without a key
- ○ **D.** To delete data after transmission

Q12 (1 mark)

How many layers does the TCP/IP model have?

- ○ **A.** Two layers
- ○ **B.** Three layers
- ○ **C.** Four layers
- ○ **D.** Seven layers

Q13 (2 marks)

A user is entering their bank details on a website.
Why should the website use HTTPS rather than HTTP?

- ○ **A.** HTTPS makes the website load faster
- ○ **B.** HTTPS allows the website to display more images
- ○ **C.** HTTPS is required for all websites regardless of content
- ○ **D.** HTTPS encrypts the data, protecting sensitive bank details from interception

Q14 (2 marks)

A web developer needs to upload website files to a remote server.
Which protocol would be most appropriate?

- ○ **A.** SMTP because it sends data
- ○ **B.** HTTP because websites use it
- ○ **C.** FTP because it is designed for transferring files
- ○ **D.** POP because it connects to servers

Q15 (2 marks)

Which combination of protocols would an email application use to send an email and then retrieve incoming emails while keeping them synchronised across multiple devices?

- ○ **A.** HTTP to send, FTP to receive
- ○ **B.** SMTP to send, IMAP to receive

- ○ **C.** FTP to send, POP to receive
- ○ **D.** IMAP to send, SMTP to receive

Q16 (2 marks)

A company wants to update their network security without replacing all their hardware.
How does the layered model help with this?

- ○ **A.** Each layer is independent, so the security layer can be updated without affecting hardware-level protocols
- ○ **B.** Layers prevent any updates from being made
- ○ **C.** All layers must be replaced together at the same time
- ○ **D.** Layers only exist in theory and have no practical benefit

Q17 (2 marks)

In the TCP/IP protocol suite, what are the different roles of TCP and IP?

- ○ **A.** TCP and IP perform identical functions
- ○ **B.** TCP handles addressing and routing; IP handles reliable data delivery
- ○ **C.** TCP handles reliable data delivery and error checking; IP handles addressing and routing
- ○ **D.** Neither TCP nor IP is used in modern networks

Q18 (2 marks)

A user reads emails on both their phone and laptop. They notice emails read on the phone still appear as unread on the laptop.
Which protocol is likely being used and why is this happening?

- ○ **A.** SMTP - it does not track read status
- ○ **B.** IMAP - it always shows emails as unread
- ○ **C.** POP - it downloads emails to each device independently without synchronisation
- ○ **D.** HTTP - it cannot handle emails properly

Q19 (2 marks)

A school buys network equipment from different manufacturers: switches from one company, routers from another, and computers from various brands.
How do network standards enable this mixed equipment to work together?

- ○ **A.** Standards force all manufacturers to make identical products
- ○ **B.** Standards define common rules that all manufacturers follow, ensuring their equipment can communicate
- ○ **C.** Standards prevent different brands from being used together
- ○ **D.** Standards are only recommendations that manufacturers can ignore

Q20 (2 marks)

An attacker intercepts network traffic between a user and an HTTPS website.
What prevents the attacker from reading the captured data?

- ○ **A.** The data is encrypted and the attacker does not have the decryption key
- ○ **B.** The data travels too fast to be read
- ○ **C.** HTTPS automatically deletes intercepted data
- ○ **D.** The attacker's computer cannot receive network data

Q21 (2 marks)

A company website displays product information to customers. Which protocol enables customers' browsers to access these web pages?

- ○ **A.** SMTP because it transfers information
- ○ **B.** FTP because websites contain files
- ○ **C.** HTTP or HTTPS because they transfer web pages from servers to browsers
- ○ **D.** POP because it retrieves data from servers

Q22 (2 marks)

In the TCP/IP model, what is the purpose of the Application layer?

- ○ **A.** To physically transmit electrical signals through cables
- ○ **B.** To route packets between different networks
- ○ **C.** To convert IP addresses into physical hardware addresses

○ **D.** To provide network services directly to user applications like web browsers and email clients

Q23 (2 marks)

When an email is sent from person A to person B, which statement about SMTP is correct?

○ **A.** SMTP only operates on the sender's computer

○ **B.** SMTP delivers the email directly to the recipient's inbox

○ **C.** SMTP is only used by the recipient to read the email

○ **D.** SMTP transfers the email from the sender to their mail server, and between mail servers until it reaches the recipient's mail server

Q24 (3 marks)

A small business is setting up their IT infrastructure. They need to: host a public website, allow employees to send and receive emails from multiple devices, and let customers download product manuals. Analyse which protocols would be needed for each requirement.

○ **A.** HTTP for everything - it is a universal protocol

○ **B.** HTTPS for the website, SMTP with IMAP for email, and FTP for file downloads

○ **C.** SMTP for the website, POP for email, and HTTP for downloads

○ **D.** FTP for everything since all requirements involve file transfer

Q25 (3 marks)

An online shop processes credit card payments. A security consultant recommends implementing HTTPS across the entire website, not just the payment pages.
Analyse why this recommendation is appropriate.

- ○ **A.** HTTPS only protects payment data, so it is unnecessary elsewhere
- ○ **B.** HTTPS on all pages protects login credentials, browsing history, and prevents attackers from intercepting session cookies that could hijack accounts
- ○ **C.** HTTPS slows down the website so much that it should only be used where legally required
- ○ **D.** Encryption is only needed when data is stored, not when transmitted

Q26 (3 marks)

A network administrator is troubleshooting why users cannot access a website. They determine that users can ping the web server's IP address successfully, but the web pages do not load.
Analyse which layer of the TCP/IP model likely has the problem and why.

- ○ **A.** The Network Access layer because cables are not connected
- ○ **B.** The Internet layer because IP routing is not working
- ○ **C.** The Application layer because lower layers work (ping succeeds) but the HTTP service is likely not responding
- ○ **D.** All layers must be faulty simultaneously

Q27 (3 marks)

A company is choosing email settings for employees who work from different locations using laptops, phones, and tablets. The IT department

is debating between POP3 and IMAP.
Analyse why IMAP would be the better choice.

- ○ **A.** POP3 is better because it downloads emails faster
- ○ **B.** POP3 provides better synchronisation across devices
- ○ **C.** IMAP only works with specific email applications
- ○ **D.** IMAP keeps emails on the server, so all devices show the same emails, read/unread status, and folder organisation - essential for multi-device mobile workers

Q28 (3 marks)

A web hosting company needs to provide complete service for a client website. Visitors should be able to view web pages securely, and the client needs to upload website files to the server.
Analyse the protocols needed for this complete solution.

- ○ **A.** Only HTTP is needed for both viewing and uploading
- ○ **B.** HTTPS for secure visitor access to web pages; FTP or SFTP for the client to upload website files to the server
- ○ **C.** SMTP for viewing pages and POP for uploading files
- ○ **D.** IMAP for viewing and HTTPS for uploading

Q29 (3 marks)

The Wi-Fi standard has evolved from 802.11b to 802.11n to 802.11ac to 802.11ax (Wi-Fi 6). Despite these changes, newer routers still work with older devices.
Analyse how standards enable this backwards compatibility.

- ○ **A.** Old devices are secretly upgraded when connecting to new

routers

- **B.** Standards define the core communication rules; newer versions add features while maintaining support for the original rules that older devices understand
- **C.** Backwards compatibility is impossible - old devices cannot work with new routers
- **D.** Manufacturers agree informally without any standards involved

Q30 (3 marks)

When downloading a software update, it is critical that every byte arrives correctly. If any data is lost or corrupted, the software might not work or could be dangerous to install.
Analyse how TCP ensures reliable delivery for this scenario.

- **A.** TCP sends data once and hopes it arrives correctly
- **B.** TCP splits data into packets with sequence numbers, confirms each packet's arrival with acknowledgements, and requests retransmission of any lost or corrupted packets
- **C.** TCP only works for small files, not software updates
- **D.** TCP relies entirely on the physical cables to prevent data loss

Total: **55** marks • Time: **30 min**

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